

ROBUST MULTI-IMAGE STITCHING

Panorama Generation with Feature Matching and Global Alignment

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A feature-based panorama generation system supporting both pairwise and unordered multi-image stitching. The pipeline combines SIFT/FLANN matching, RANSAC homography estimation, graph-based global alignment, perspective warping, and weighted blending.

1. Project Overview

The project addresses automatic panorama construction from overlapping photographs. Rather than treating stitching as a fixed sequence of image pairs, the multi-image system first discovers which images are geometrically related, then aligns the connected set in a shared coordinate frame.

The implementation was designed around two goals: reliable geometric registration and tolerance to unordered input. A pairwise pipeline validates the low-level vision components, while the multi-image pipeline extends them into a graph-based global system.

2. System Design

2.1 Pairwise Registration

- SIFT is used to detect local keypoints and compute descriptors.
- FLANN performs nearest-neighbor descriptor matching, followed by Lowe's ratio test to reject ambiguous correspondences.
- A 3 x 3 projective homography is estimated with RANSAC using a 5-pixel reprojection threshold.
- The transformed image is warped into a common canvas and overlapping pixels are blended.

2.2 Multi-Image Alignment

For an unordered image collection, each image is represented as a graph node. An edge is added when a pair produces a valid homography with at least 15 RANSAC inliers. The image with the largest graph degree is selected as the reference frame.

Global transformations are propagated from the reference image through graph traversal. This removes the need for the input files to be pre-sorted and allows images without direct overlap with the reference frame to be aligned through intermediate views.

3. Engineering Decisions

- Homography validation rejects transformations with unreasonable determinant, scale, or perspective terms.
- Canvas bounds are derived from transformed image corners before global warping.
- Size guards prevent abnormal homographies from allocating excessively large canvases.
- Weighted blending gives central image regions slightly higher contribution than border regions, reducing hard seams.

4. Evaluation Summary

Metric	Observed Result
Average feature matches	1500+
RANSAC inlier ratio	~60%
Pairwise stitching time	< 0.5 s
Alignment error	< 3 px

The system successfully handled the provided pairwise and multi-view test sets, including image groups ranging from 2 to 6 views. The graph structure remained effective even for partially connected cases, where only a subset of all possible image pairs overlapped.

5. Strengths and Limitations

Strengths. The pipeline is modular, interpretable, and robust to unordered input. Graph-based alignment makes the system more flexible than simple sequential stitching, while explicit homography validation improves stability.

Limitations. The current blending strategy is intentionally lightweight and does not address exposure differences, parallax, or complex seam placement. Large viewpoint changes can also reduce the reliability of a single-homography model.

6. Future Improvements

- Multi-band blending and seam optimization for smoother transitions.
- Exposure compensation for images captured under different lighting conditions.
- Bundle adjustment for globally optimized camera alignment.
- Cylindrical or spherical projection for wider panoramas.
- GPU acceleration for feature extraction, warping, and blending.

7. Repository

Source code and report: github.com/jhzheng-241220179/multi-image-stitching